

Acne Severity Assessment in Facial Images

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Introduction

- Acne assessment is currently subjective and inconsistent.
 - Manual lesion counting is time-consuming
 - High inter-observer variability
 - Need for objective AI support



Hayashi Criterion

- Hayashi Criterion (2008):** Acne severity is determined by inflammatory lesion count.

Hayashi Severity Grading (2008)

Lesion Count	Severity Grade
0 - 5	Mild
6 - 20	Moderate
21 - 50	Severe
> 50	Very Severe

Adapted from Hayashi et al., 2008

- Our framework leverages this definition by learning the lesion count–severity correlation.

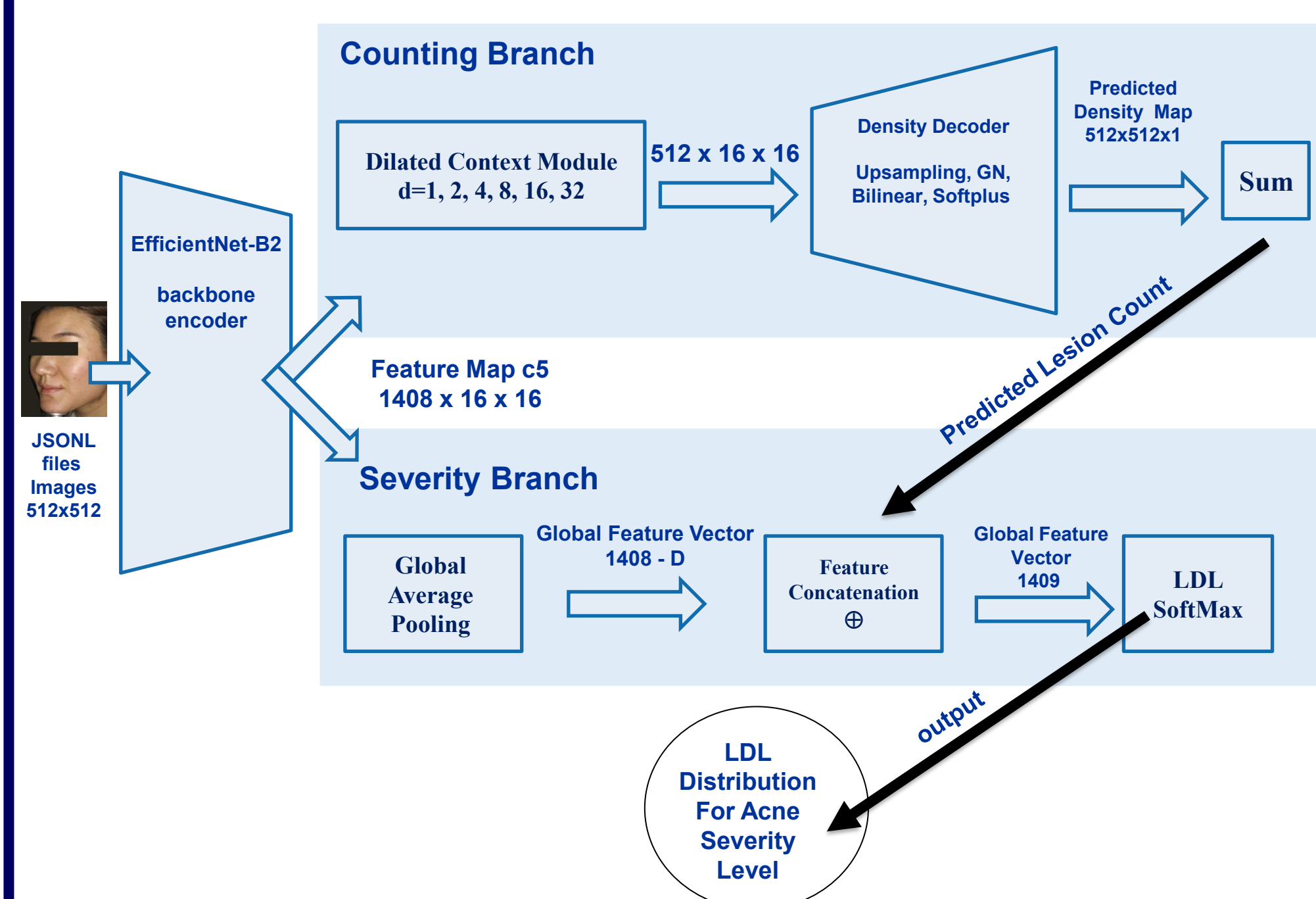
Prior Work

- Wu et al. (2019):**
 - Introduced Label Distribution Learning (LDL) for acne severity grading, representing severity as a probability distribution rather than a single hard label. This captures clinical ambiguity in borderline cases but makes limited use of explicit spatial lesion information.
- Alzahrani et al. (2022):**
 - Proposed an attention-guided approach for exploiting the spatial distribution of lesions and improving lesion analysis in dense or small-lesion regions. However, the link between lesion analysis and probabilistic severity grading remains limited.

Goals

- Multi-task system: count + severity grading.
- Use clinical correlation.
- Model uncertainty with LDL.

AcneFusionNet Unified Multi-Task Framework



AcneFusionNet architecture integrating lesion counting and severity grading

- Shared Backbone**
 - EfficientNet-B2 extracts multi-scale features.
- Counting Branch**
 - Generates a continuous pixel-wise density map, where the integral yields the total lesion count ($\hat{C}$) inspired by spatial lesion analysis approaches (Alzahrani et al., 2022).
- Feature Fusion Mechanism**
 - Predicted count ($\hat{C}$) is injected into the severity classification branch.
 - Fusion Vector:

$$f_{\text{fusion}} = [f_{\text{global}}, \hat{C}]$$
 Fusion mirrors clinical reasoning where lesion quantity informs severity.
- Severity Branch (LDL)**
 - Predicts a Label Distribution (probability vector) over the 4 Hayashi grades, inspired by Wu et al. (2019).

Multi-Task Loss Function

- We define a joint loss function that combines regression and classification objectives:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{count}} + \lambda \cdot \mathcal{L}_{\text{severity}}, \quad \lambda = 7$$

- $\mathcal{L}_{\text{count}}$: Mean Squared Error used for the pixel-wise density map regression, ensuring accurate lesion counting.

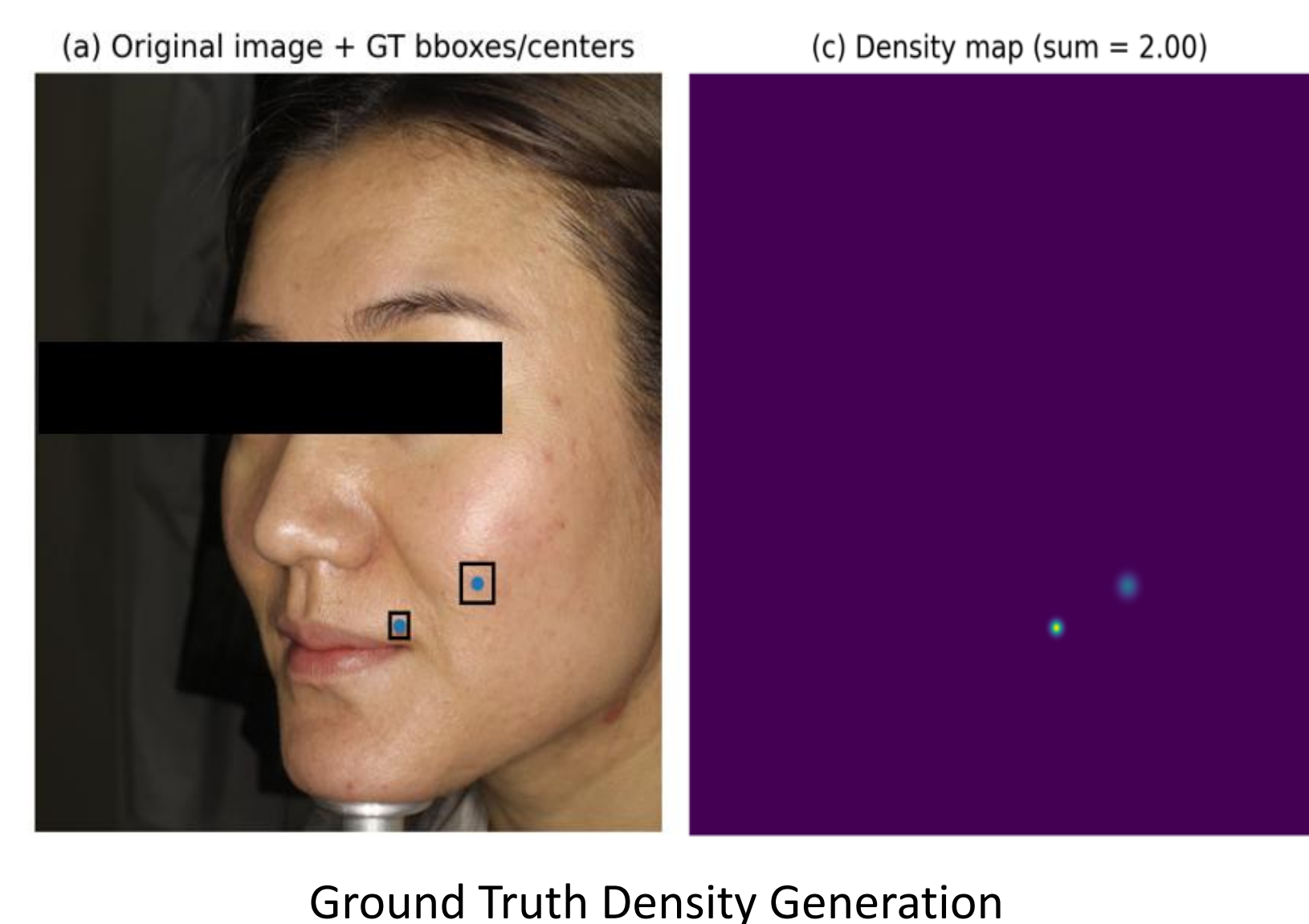
$$\mathcal{L}_{\text{count}}(D_{\text{pred}}, D_{\text{GT}}) = \frac{1}{N} \sum_{i=1}^N (D_{\text{pred}}^{(i)} - D_{\text{GT}}^{(i)})^2$$

- $\mathcal{L}_{\text{severity}}$: Weighted Kullback-Leibler Divergence used to align the predicted label distribution with the clinical ground truth.

$$\mathcal{L}_{\text{severity}} = \sum_{k=0}^{K-1} w_k \cdot L_{GT,k} \cdot \ln \left(\frac{L_{GT,k}}{\hat{L}_k} \right)$$

Dataset

- ACNE04 Dataset: 1,457 facial images.
- Bounding boxes + severity labels.
- Evaluation: 5-Fold Cross-Validation (80% Train/Val, 20% Test per fold)



Results

- Evaluation on ACNE04 using 5-Fold Cross-Validation.

Comparison with prior methods

Model	Method	Accuracy
YOLOv3	Detection	63.70%
ResNet-50	Classification	78.42%
Wu et al. (2019)	LDL + Counting	84.11%
AcneFusionNet (Ours)	Fusion + LDL	85.86%
Alzahrani (2022)	Attention + Faster R-CNN	91.5%

- Mean classification accuracy of 85.86%.
- Improves over Wu et al. (2019) LDL baseline by +1.75%.
- Provides a unified framework for lesion counting and severity grading

Conclusions

- Unified Framework:**
 - We combine these two complementary ideas in a unified framework: density-based lesion counting informs an LDL severity grading branch, enabling both spatially aware lesion analysis and probabilistic clinical grading.
- Objective Assessments:**
 - The system provides a consistent and objective alternative to subjective manual grading.
- Robustness:**
 - The integration of density regression allows for accurate counting even in dense lesion clusters where standard detectors fail.

Future Work

- Semantic Classification:** Extend the detection branch to distinguish between lesion types (comedones, papules, pustules) to support GAGS scoring.
- Longitudinal Analysis:** Develop a framework to track patient progress over time using sequential images.